

# Mehdy Mokhtari

Machine Learning Engineer — AI Engineer

+98-936-017-2983 | [mh.mokhtari7@gmail.com](mailto:mh.mokhtari7@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## SUMMARY

Final-year Computer Engineering student (GPA: **18.21**/20, Konkoor: **68th**) with hands-on experience developing machine learning, deep learning, computer vision, NLP, LLM, and RAG systems through industry internships, research, and end-to-end projects. Experienced in designing, training, evaluating, and deploying AI solutions using PyTorch and modern ML frameworks. Seeking a Machine Learning Engineer or AI Engineer role to build production-ready intelligent systems and grow as an engineer.

## EDUCATION

### University of Tehran

Tehran, Iran

*B.Sc. in Computer Engineering*

2022 – 2026

- **GPA:** 18.21/20
- **Relevant Coursework:** Artificial Intelligence, Data Science, NLP, LLM, Trustworthy AI

### Atomic Energy High School (NODET)

Tehran, Iran

*Diploma in Mathematics & Physics*

2019 – 2022

- **GPA:** 19.76/20
- **Konkoor:** Ranked 68th nationwide

## EXPERIENCE

### ML Engineer – Distributed Acoustic Sensing (DAS) Security System

Fall 2025 (3 Months)

*Farhikhtegan Elm o Sanaat Inc.*

Tehran, Iran

- Independently developed the AI proof-of-concept for a Distributed Acoustic Sensing (DAS) security system to classify infrastructure events, including human intrusion, excavation, heavy machinery activity, and background signals.
- Designed the end-to-end preprocessing pipeline, including FFT, denoising, spectrogram generation, data augmentation, and class balancing for time-series and vision-based inputs.
- Implemented and evaluated multimodal deep learning architectures combining LSTMs, Transformers, and pretrained CNNs for event classification from DAS signals and spectrograms.
- Optimized model architectures and inference pipelines for CPU-based edge deployment under latency and computational constraints.

### ML Engineer Intern – Handwritten Sequence OCR

Summer 2025 (1 Month)

*Roshan Inc.*

Tehran, Iran

- Developed deep learning models in PyTorch to improve the company's handwritten Persian digit sequence OCR system using the ORAND-CAR dataset.
- Implemented and trained three OCR approaches: traditional image processing, CNN-BiLSTM (CTC), and Transformer-based attention.
- Improved sequence recognition accuracy from 47% to 87% and character-level accuracy from 79% to 91%.
- Optimized disk data loading with parallel I/O, reducing training and evaluation time.
- Ranked among the top 5 of 50 interns based on technical evaluation and OCR model performance.

### ML Engineer Intern – Persian Question Answering (ParsQA)

Summer 2025 (1 Month)

*Roshan Inc.*

Tehran, Iran

- Developed an end-to-end Persian extractive QA system using Transformer models (ParsBERT, XLM-RoBERTa-Large) in a SQuAD-style pipeline
- Implemented data preprocessing, tokenization, and answer span alignment for Persian text, addressing RTL-specific NLP challenges and normalization issues
- Fine-tuned and evaluated models using EM and F1 metrics, performing error analysis and optimizing training under GPU memory constraints

### Research Assistant – Causality-Aware Knowledge Editing in LLMs

2026 – Present

*University of Tehran (Supervisor: Dr. Yaghoubzade, PhD Student: Saleh)*

Tehran, Iran

- Developing evaluation pipelines to analyze LLM behavior under targeted edits

## PROJECTS

---

### **Speech Emotion Recognition (SER) System** | *ML, DL, Feature Engineering, Docker, Power BI* | [GitHub](#)

- Developed a multi-dataset SER system combining 4 datasets with engineered acoustic features stored in MySQL
- Applied statistical feature selection (Information Gain, Correlation) and trained traditional ML and deep learning models, evaluated with accuracy, F1, precision, and confusion matrices
- Built end-to-end workflow with Python, Docker deployment, and Power BI reporting for interactive visualization of results

### **Real-Time Payment Data Pipeline** | *Kafka, PySpark, Spark Streaming, Python* | [GitHub](#)

- Built a high-throughput real-time payment data pipeline, streaming transactions via Kafka with error logging and fraud detection alerts
- Processed historical data with PySpark, analyzed transaction patterns, and visualized business insights using Python dashboards

### **Satellite-to-Map Image Translation using GANs** | *Deep Learning, Computer Vision, PyTorch* | [GitHub](#)

- Developed a conditional GAN (pix2pix) with U-Net generator and PatchGAN discriminator for image translation
- Optimized model performance using PSNR/SSIM metrics, iterative hyperparameter tuning, and logging

### **Legal Document RAG Agent** | *LangGraph, LangChain, LanceDB, RAGAS, Chainlit*

- Built an agentic legal RAG system using LangGraph with query rewriting, intent classification, metadata extraction, filtered LanceDB retrieval, reranking, and LLM-based answer generation for multilingual legal documents.
- Processed scanned PDFs with OCR, indexed documents using metadata-aware vector search, evaluated answer faithfulness and relevance with RAGAS, and profiled node-level latency to identify pipeline bottlenecks.

### **Trustworthy Machine Learning** | *Generalization, Robustness, Interpretability, Fairness, PyTorch* | [GitHub](#)

- Implemented and evaluated trustworthy ML techniques including Just Train Twice (JTT) for mitigating spurious correlations on Colored MNIST, improving out-of-distribution generalization under distribution shift.
- Evaluated model robustness and interpretability by implementing FGSM and PGD adversarial attacks alongside LIME and Grad-CAM explanations for CNN image classifiers.

### **Agentic Travel Planning Assistant** | *LangGraph, LangChain, LanceDB, RAG, ReAct, LLMs*

- Developed a ReAct-based travel planning agent using LangGraph with tool routing across Amadeus APIs, Tavily Search, semantic FAQ retrieval, weather, currency conversion, flight/hotel search, and itinerary generation.
- Built a RAG pipeline using LanceDB and BAAI/bge-small-en-v1.5 embeddings over travel guides and FAQs, integrating LlamaParse for document ingestion and metadata-aware semantic retrieval to support trip planning.

### **Soccer Player Segmentation** | *Computer Vision, Image Segmentation, Clustering* | [GitHub](#)

- Developed an unsupervised player segmentation pipeline using engineered pixel features, clustering, cluster merging, and post-processing to generate player masks, evaluated with Dice and IoU metrics.

## TECHNICAL SKILLS

---

**Languages:** Python, SQL, C++

**Frameworks & Libraries:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, SentenceTransformers, OpenCV, YOLO, MLflow

**LLM & Generative AI:** LangChain, Ollama, RAG, Prompt Engineering, Fine-Tuning (LoRA/QLoRA), Quantization, Pruning, LLMs (GPT, T5, Qwen, TinyLlama)

**MLOps & Deployment:** FastAPI, Flask, Docker, Streamlit, Gradio, TorchServe

**ML Techniques:** CNN/RNN/LSTM/Transformer, GAN/VAE/Autoencoder, XGBoost/LightGBM/Random Forest, Clustering, PCA, Active Learning, Adversarial Robustness, Edge AI, Knowledge Distillation

**Data & Infrastructure:** Spark, Kafka, Git, Jupyter Notebook, Google Colab, Kaggle, MySQL, PostgreSQL, MongoDB

**Visualization & BI:** Matplotlib, Seaborn, Plotly, Power BI, Tableau

## RELEVANT COURSEWORK

---

**Advanced Python Programming:** Quera

**Generative AI:** Daneshkar

**Database (SQL):** Quera

**MLOps:** Hamrah.academy

**Linux:** Quera